

Session 4 — Project Tasks — P4A 2026

Complete these tasks in your group notebook before Session 5. Bring any questions to the start of Session 5.

What you covered today

- DuckDB: running SQL queries directly on Pandas DataFrames
- Matplotlib and Seaborn: histograms, bar charts, heatmaps
- Saving results: writing DataFrames to CSV and JSON
- Python modules: moving functions into a `.py` file and importing them into a notebook

Tasks

1. Query with DuckDB — postings by industry

Load the `jobs_job_industries` and `mappings_industries` tables from the database into DataFrames. Using DuckDB, write a SQL query that JOINS these two tables with your full-time postings from `postings_with_benefits` and returns the **number of full-time postings per industry name**, ordered from most to fewest. Load the result into a new DataFrame.

Write one sentence on why you used DuckDB here instead of a Pandas merge.

2. Analytics challenge — benefits by industry

Using DuckDB, JOIN `postings_with_benefits` with `jobs_job_industries` and `mappings_industries`. Filter to full-time postings where the `benefits` column contains `'Medical insurance'`. Return the **number of matching postings per industry name**, ordered from most to fewest. Limit to the top 10.

Write one sentence: does the industry ranking for medical insurance coverage match the overall posting ranking from Task 1, or are there industries that stand out?

3. Visualise postings by industry

Using the result from Task 1, create a **horizontal bar chart** showing the top 15 industries by number of full-time postings. The industry with the most postings should appear at the top. Label both axes and give the chart a title.

4. Visualise the distribution of views

Using the full-time postings, create a **histogram** of the `views` column. Choose a sensible number of bins. Add a vertical line marking the median number of views. Does the distribution look symmetric, or is it skewed? Write one sentence describing what you see.

5. Save your results

Save two outputs from your analysis:

- The industry posting counts (from Task 1) as a **CSV file** in your project's **output/** folder.
- The top 10 countries from Session 3, Task 8 as a **JSON file** in your project's **output/** folder.

Name the files clearly so they are self-explanatory.

6. Create **analysis.py**

Today you learned how to move functions out of a notebook and into a **.py** module. Now apply that to your project.

Create a file called **analysis.py** in your project folder and add all four required functions:

- **load_postings(engine)** — loads the **postings_with_benefits** table from the database and returns a DataFrame. Parse **listed_time** from Unix milliseconds to UTC datetime.
- **filter_fulltime(df)** — filters to rows where **formatted_work_type == 'Full-time'** and returns a copy.
- **company_summary(engine)** — loads the **companies_companies** table, filters to US companies, and returns a DataFrame. This is the logic from Session 3, Tasks 3 and 5.
- **top_industries(df_postings, df_job_industries, df_industries, n)** — returns the top N industries by posting count as a DataFrame. This is the DuckDB result from Task 1, expressed as a reusable function.

Each function must have a docstring. In your notebook, import and call all four to confirm they produce the same results as before.

Bring to Session 5

- Task 1: which industries dominate the job market in this dataset? Does this match what you expected?
- Task 2: which industries stand out for medical insurance coverage relative to their overall posting volume?
- Task 4: the views distribution is likely right-skewed. What kinds of postings might attract many more views than average? Is that useful signal or noise?